

1. Empirical inputs

abundance / taxon-function
table + group labels +
reference tree



2. Tree perturbation (optional)

branch-length jitter +
nearest-neighbor
interchange



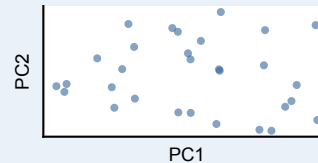
3. Synthetic pools + effect modulation

PCAM | MDC-TF-MC
bidirectional realized ω^2
gradient



4. Distance recompute + bootstrap PERMANOVA

Gemelli | PhyloFunc
B resampling iterations,
 $\alpha = 0.05$



5. Power curves & minimum n

